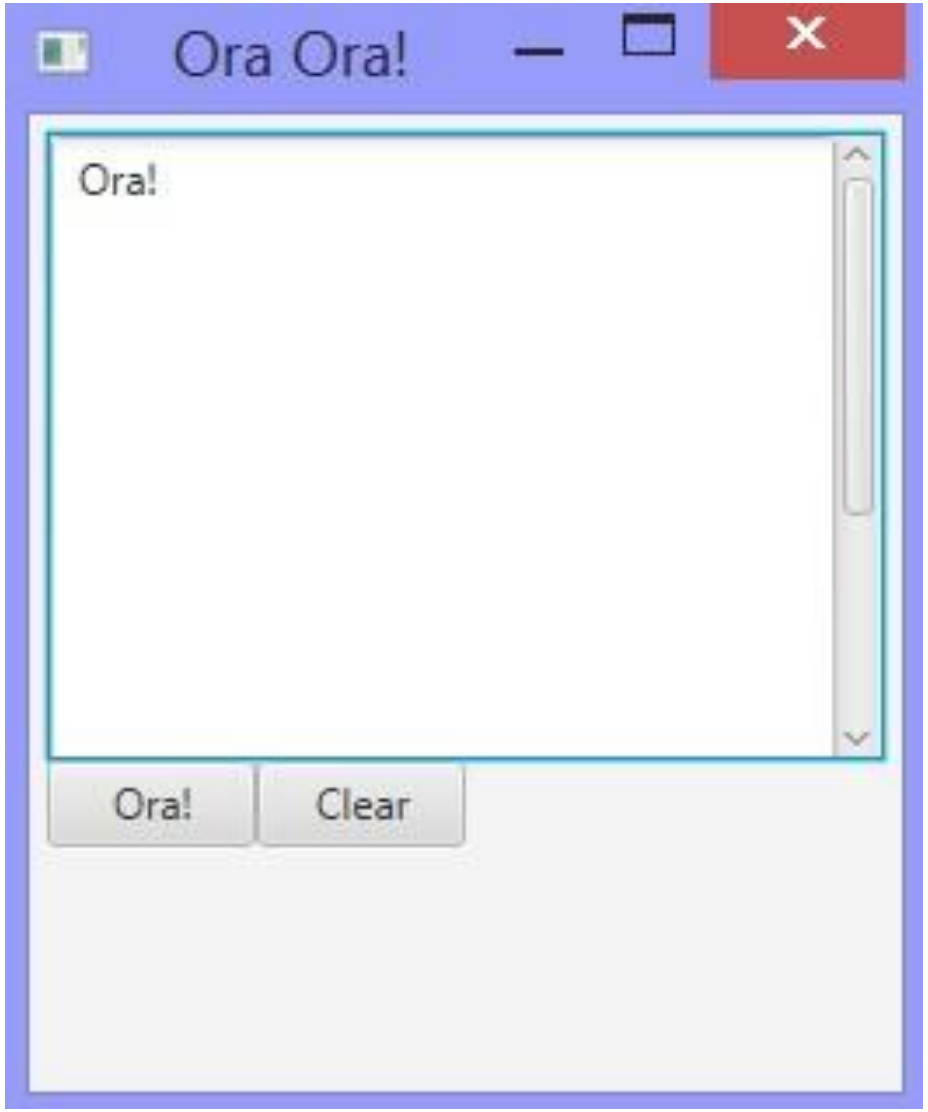
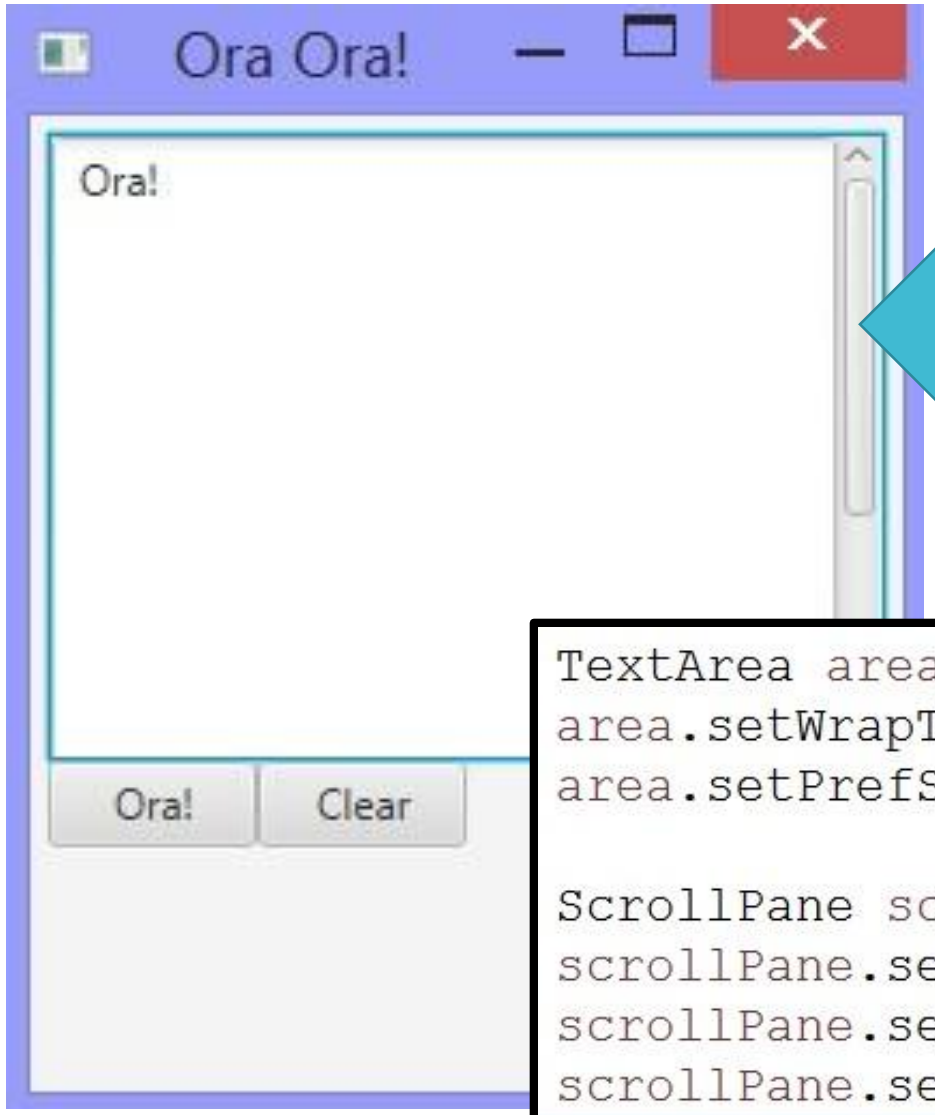
More User Interface Examples

Trying Key press and Mouse click [KeyPress.java]



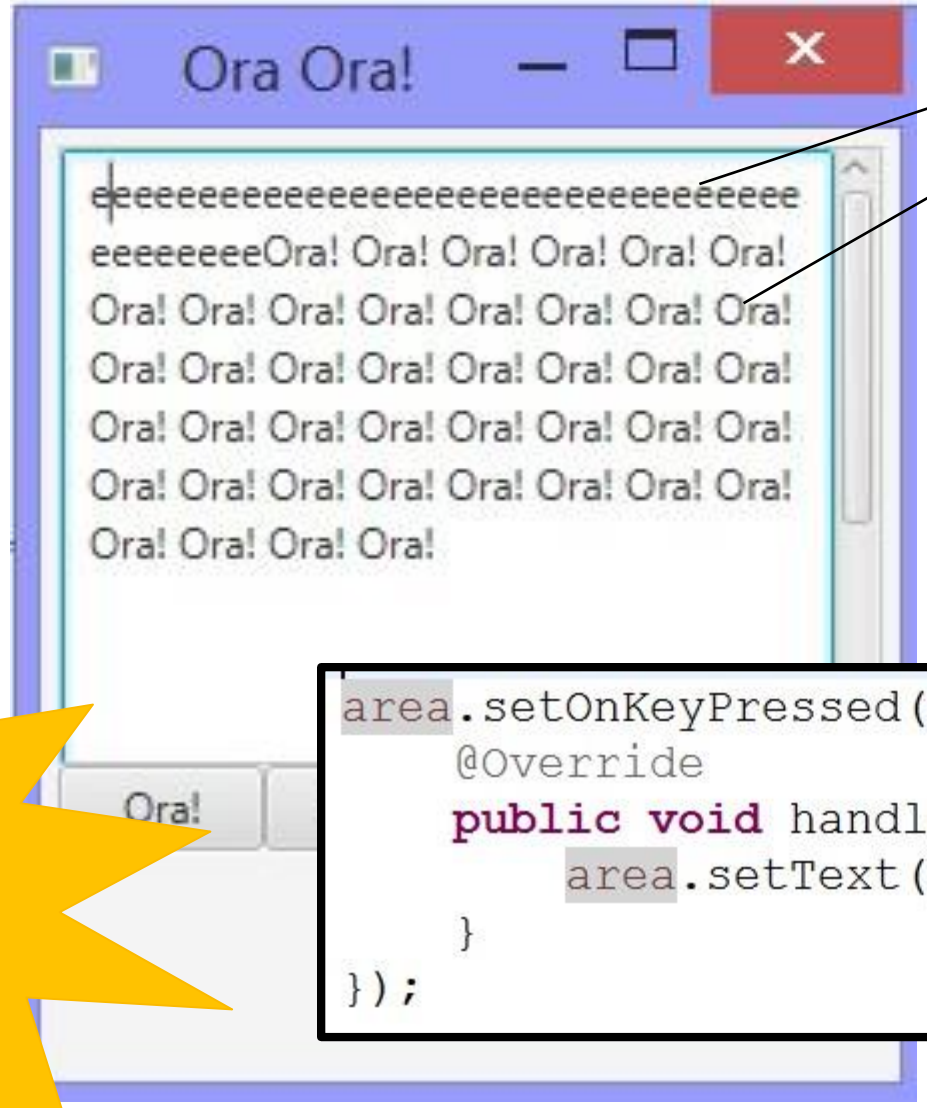
Interest Points: Text Wrap and Scroll Pane



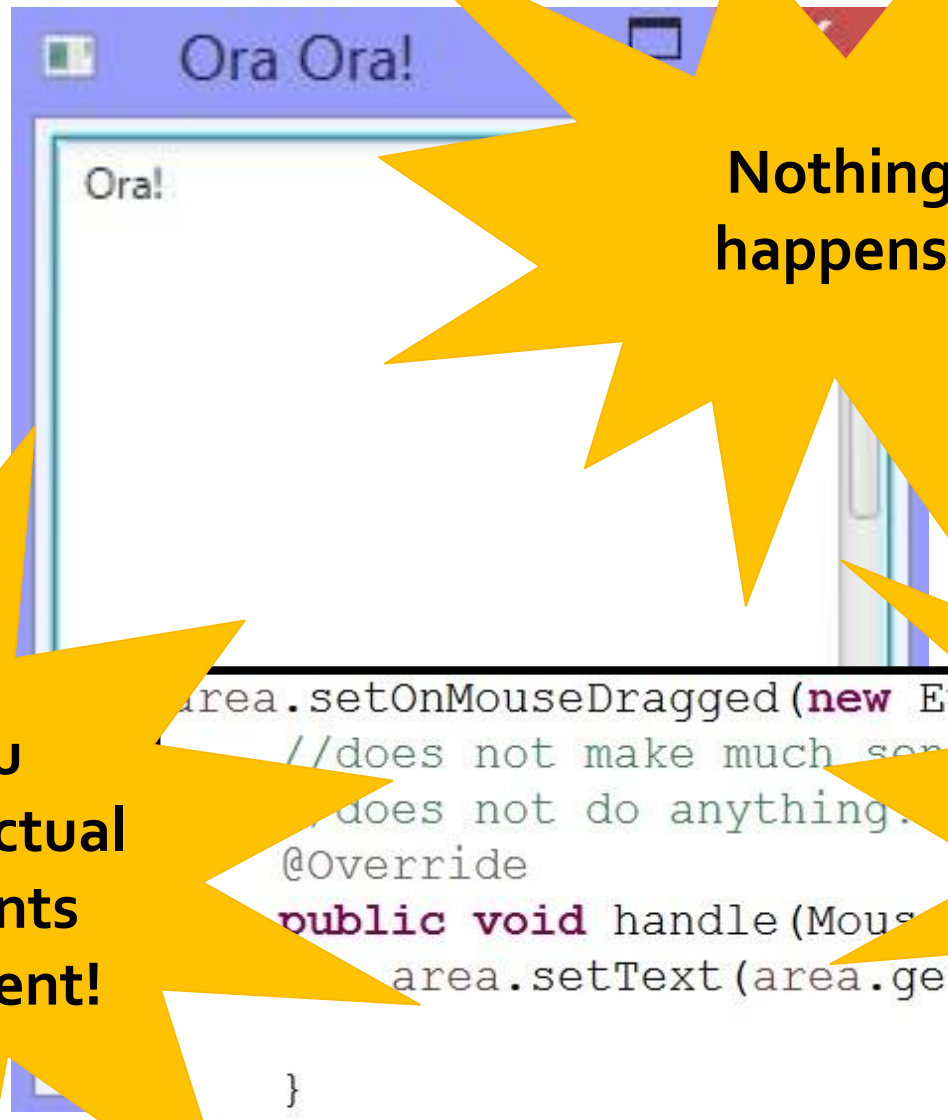
Scroll Pane

```
TextArea area = new TextArea();
area.setWrapText(true); // important prop
area.setPrefSize(250, 280);

ScrollPane scrollPane = new ScrollPane();
scrollPane.setContent(area);
scrollPane.setFitToWidth(true);
scrollPane.setPrefWidth(200);
scrollPane.setPrefHeight(180);
```

KeyPress in TextArea


Ora Ora!
eeee

Mouse Drag in TextArea!



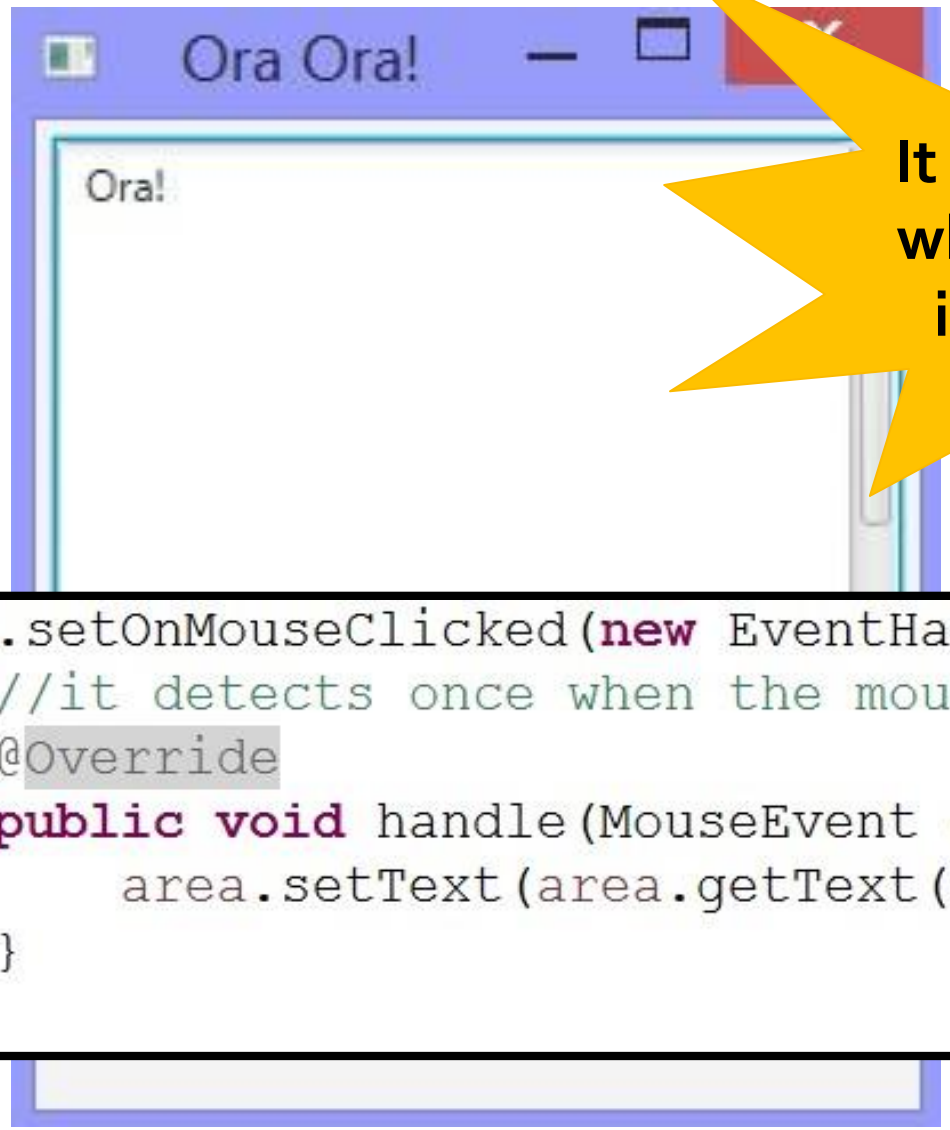

```
area.setOnMouseDragged(new Eve
    //does not make much se
    //does not do anything.
    @Override
    public void handle(Mous
        area.setText(area.getText

    }
};
```

Nothing happens!

Make sure you understand the actual meaning of events inside a component!

But it works when you are dragging the left most side of the area!

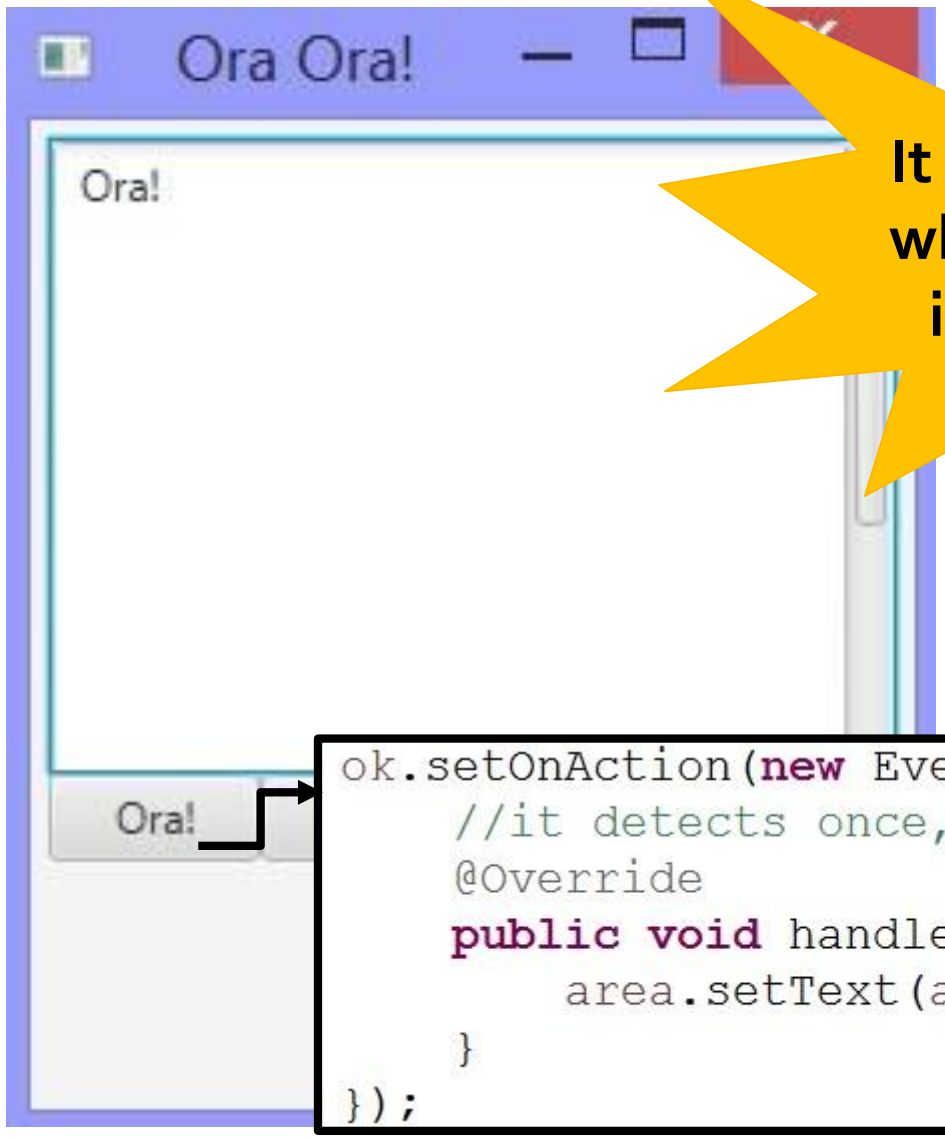
Mouse Click in TextArea



It detects ONCE, when the mouse is RELEASED.

```
area.setOnMouseClicked(new EventHandler<MouseEvent>() {
    //it detects once when the mouse is released.
    @Override
    public void handle(MouseEvent event) {
        area.setText(area.getText()+ "Ora! ");
    }
});
```

Button click



It detects ONCE, when the button is RELEASED.

```
ok.setOnAction(new EventHandler<ActionEvent>() {
    //it detects once, when you release!
    @Override
    public void handle(ActionEvent event) {
        area.setText(area.getText()+ "Ora! ");
    }
});
```

KeyPress in TextArea, how to fire the action only once?

Use boolean condition to check?

Set the boolean on key press and reset it on key release

[KeyTrigger.java]

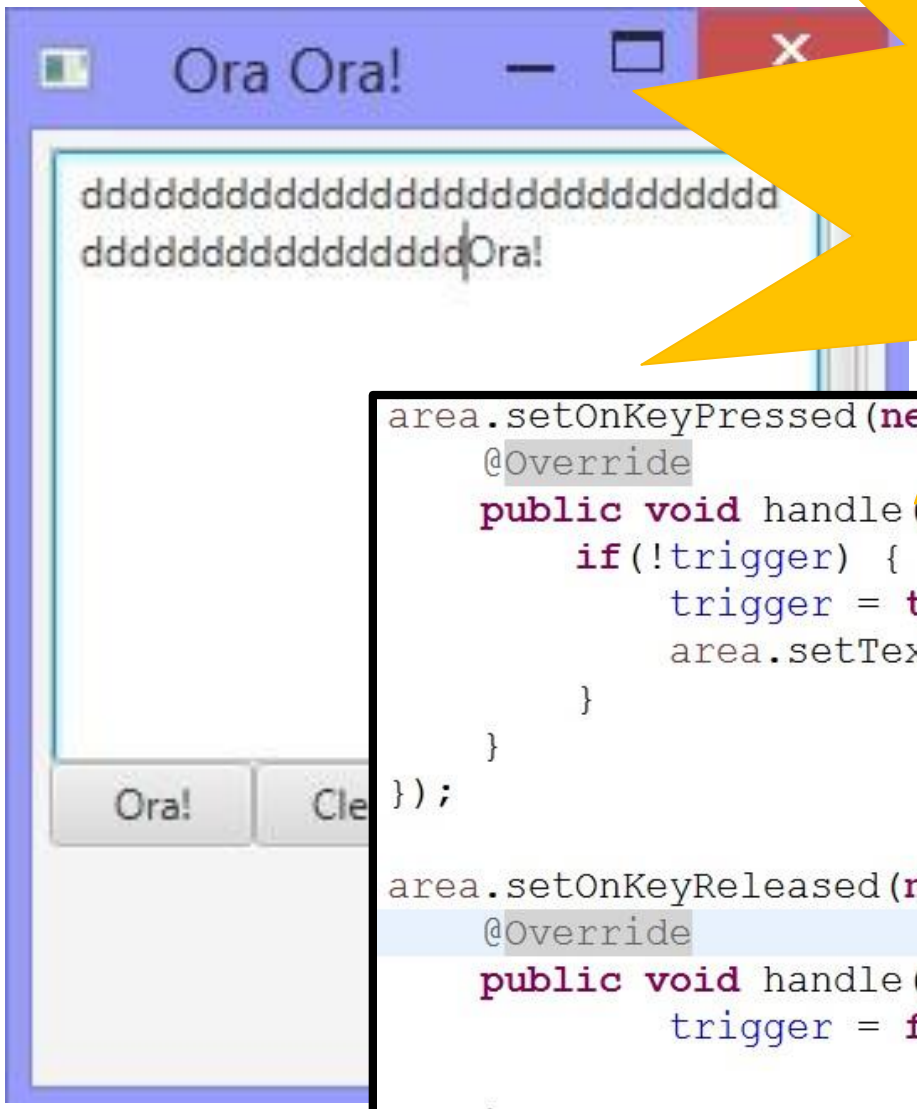
```
boolean trigger = false;
area.setOnKeyPressed(new EventHandler<KeyEvent>() {
    @Override
    public void handle(KeyEvent event) {
        if(!trigger) {
            trigger = true;
            area.setText(area.getText()+ "Ora! ");
        }
    }
});
```

But it's a compile error!!

Cannot use local variable!

Solution 1: use a class variable instead

[KeyTrigger01.java]




```
area.setOnKeyPressed(new
    @Override
    public void handle(KeyEvent
        if(!trigger) {
            trigger = true;
            area.setText(area.ge
        }
    }
});

area.setOnKeyReleased(new EventHandler<KeyEvent>() {
    @Override
    public void handle(KeyEvent event) {
            trigger = false;

    }
});
```

Only 1 Ora! while we keep pressing key "d". But "d" is still increasing since the text area is editable.

Solution 2: use a listener that remember the trigger status

[KeyTrigger02.java]

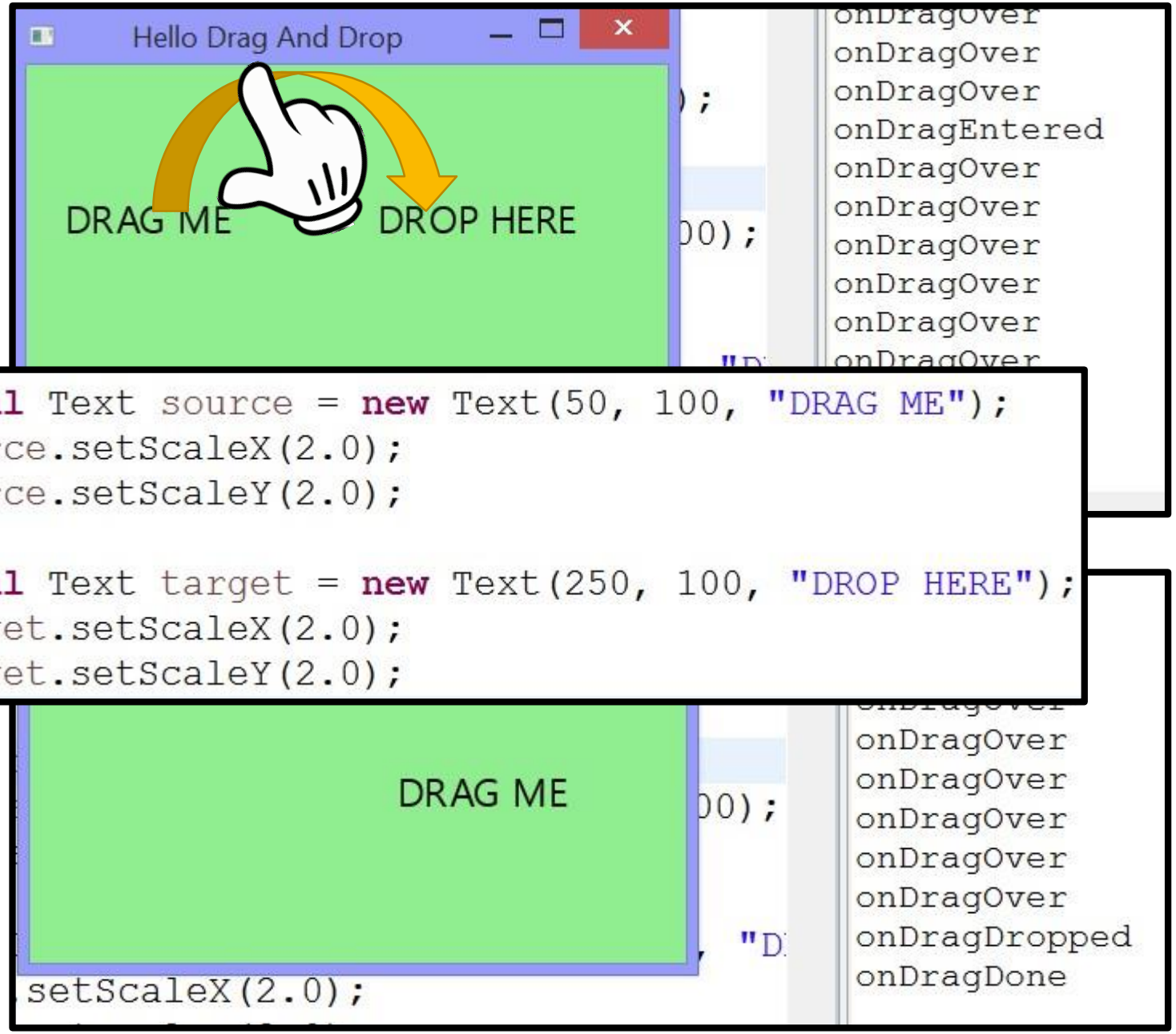
```java
MyKeyHandler areaKeyHandler = new MyKeyHandler(area);
area.setOnKeyPressed(areaKeyHandler);

area.setOnKeyReleased(new EventHandler<KeyEvent>() {
    @Override
    public void handle(KeyEvent event) {
        areaKeyHandler.setTrigger(false);
    }
});
```

```java
public class MyKeyHandler implements EventHandler{
    boolean trigger = false;
    TextArea area;
    public MyKeyHandler(TextArea a) {

    public void setTrigger(boolean t) {
        trigger =t;
    }

    @Override
    public void handle(Event event) {
        if(trigger) {
            //do nothing
        }else {
            trigger = true;
            area.setText(area.getText()+ "Ora! ");
        }
```

Yes. You can write your own handler (implements from an existing one).

Interesting feature: Drag and Drop

[HelloDragAndDrop.java] – from oracle tutorial




```
final Text source = new Text(50, 100, "DRAG ME");
source.setScaleX(2.0);
source.setScaleY(2.0);

final Text target = new Text(250, 100, "DROP HERE");
target.setScaleX(2.0);
target.setScaleY(2.0);
```

What you need to do.

Source

setOnDragDetected
- Call method startDragAndDrop and indicate its transfer mode (COPY,MOVE,or LINK)
- Copy data to transfer onto dragboard (clipboard for drag-and-drop)

setOnDragDone
- Clear original data if necessary.

Destination

setOnDragOver
- Call method acceptTransferModes indicating its transfer mode (COPY,MOVE,or LINK). The transfer mode must match the source.

setOnDragEntered

setOnDragExited

setOnDragDropped
- Call method setDropCompleted

Visual feedback

Source: setOnDragDetected to initialize.

```java
source.setOnDragDetected(new EventHandler<MouseEvent>() {
    @Override
    public void handle(MouseEvent event) {
        /* drag was detected, start drag-and-drop gesture */
        System.out.println("onDragDetected");

        /* allow MOVE transfer mode */
        Dragboard db = source.startDragAndDrop(TransferMode.MOVE);

        /* put a string on dragboard */
        ClipboardContent content = new ClipboardContent();
        content.putString(source.getText());
        db.setContent(content);

        event.consume();
    }
});
```

Copy content as a ClipBoardContent and put it in our drag board.

Consume event so that it does not get passed on to other related nodes.

Destination: setOnDragOver to set a matching transfer mode.

```java
target.setOnDragOver(new EventHandler<DragEvent>() {
    @Override
    public void handle(DragEvent event) {
        /* data is dragged over the target */
        System.out.println("onDragOver");

        /*
         * accept it only if it is not dragged from
         * the same node and if it has a string
         * data
         */
        if (event.getGestureSource() != target && event.getDragboard().hasString()) {
            /* allow for moving */
            event.acceptTransferModes(TransferMode.MOVE);
        }

        event.consume();
    }
});
```

Destination: setOnDragEntered and setOnDragExited to provide Visual Feedback

Source must not come from itself.

```
target.setOnDragEntered(new EventHandler<DragEvent>() {
    @Override
    public void handle(DragEvent event) {
        /* the drag-and-drop gesture entered the target */
        System.out.println("onDragEntered");
        /* show to the user that it is an actual gesture target */
        if (event.getGestureSource() != target &&
                event.getDragboard().hasString()) {
            target.setFill(Color.GREEN);
        }

        event.consume();
    }
});
```

Must verify that data has a correct type.

```
target.setOnDragExited(new EventHandler<DragEvent>() {
    @Override
    public void handle(DragEvent event) {
        /* mouse moved away, remove the graphical cues */
        target.setFill(Color.BLACK);
        event.consume();
    }
});
```

Destination:
setOnDragDropped

This is what happens when the mouse is released on the destination (DRAG_OVER must work and transfer mode must also be compatible).

```
target.setOnDragDropped(new EventHandler<DragEvent>() {
    @Override
    public void handle(DragEvent event) {
        /* data dropped */
        System.out.println("onDragDropped");
        /* if there is a string data on dragboard, read it and use it */
        Dragboard db = event.getDragboard();
        boolean success = false;
        if (db.hasString()) {
            target.setText(db.getString());
            success = true;
        }
        /*
         * let the source know whether the string
         * was successfully transferred and used
         */
        event.setDropCompleted(success);
        event.consume();
    }
});
```

Must be called, or the drag is never success.

Source: setOnDragDone

```
source.setOnDragDone(new EventHandler<DragEvent>() {
    @Override
    public void handle(DragEvent event) {
        /* the drag-and-drop gesture ended */
        System.out.println("onDragDone");
        /* if the data was successfully moved, clear it */
        if (event.getTransferMode() == TransferMode.MOVE) {
            source.setText("");
        }

        event.consume();
    }
});
```

If this is null, then the drag fails.